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Stability and migration of charged oxygen interstitials in ThO_2 and CeO_2

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Abstract

Density functional theory calculations have been carried out to study the stability and migration of charged oxygen interstitials in ThO_2 and CeO_2. The calculations demonstrate that the oxygen interstitial is likely to lose electrons under *p*-type conditions and gain electrons under *n*-type conditions. Neutral $\left(O^0_{split}\right)$ and singly positive $\left(O^+_{split}\right)$ O–O⟨1 1 0⟩ split interstitials, and doubly negative octahedral ($O^{2-}_{octa.}$) oxygen interstitial are found to be the lowest-energy configurations within a certain Fermi energy range. In both oxides, the O^0_{split} is the most mobile, and the migration energies of the split oxygen interstitials in ThO_2 are lower than in CeO_2, indicating higher oxygen interstitial mobility in ThO_2 than in CeO_2.
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Keywords: DFT; Charged defect; Oxygen interstitial; Stability and migration

1. Introduction

Fluorite-structured oxides such as UO_2, PuO_2 and ThO_2 are important nuclear fuels in fission reactors [1,2], where they are exposed to irradiation damage from neutrons, high-energy charged particles (e.g. fission products) and alpha decay. Such irradiation damage produces atomic point defects, as well as electrons and holes that can trap point defects. The point defects contribute to the microstructural evolution in nuclear fuel, although many of these defects recombine directly at elevated temperatures [3]. The mobility of these primary radiation defects plays an important role in the microstructural evolution, and secondary diffusion-controlled reactions between defects can result in the formation of defect clusters, dislocation loops and voids. For example, the so-called Willis cluster that is formed by oxygen interstitial aggregation has been observed in UO_2 experimentally [4]. Oxygen interstitial dislocation loops lying on the {1 1 1} plane have been observed by Yasunaga et al. [5] in CeO_2 irradiated with an electron beam. The formation of such defects may ultimately lead to the degradation of thermal and mechanical properties of nuclear materials [6]. In addition to nuclear applications, oxygen interstitial defects have been suggested to provide oxygen mobility that facilitates the oxygen storage capacity of CeO_2 in catalytic converters [3,7]. Also, in systems such as ThO_{2+x}, the anion-to-cation ratio has a strong effect on the physical and chemical properties, e.g. thermal conductivity [8]. Understanding the stability and migration of oxygen interstitials is thus of critical importance for the evaluation of material properties and performance.

In the past few decades, extensive studies of oxygen interstitial formation and transport in UO_2 have been carried out [1,9–13]. ThO_2 and CeO_2, on the other hand, have received relatively less attention [1,14,15]. In these studies, the point defects were treated as neutral charge states. However, it is well known that point defects in semiconductors and ceramics may be electrically charged due to doping, impurities or nonstoichiometry. For example, the oxygen interstitial in $PuO_{2.25}$ is found to be singly charged (O^-) [16]. Under irradiation, atomic defects are produced by displaced atoms and even greater numbers of electrons

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and holes are created by ionization. Thus, even if neutral defects are created under irradiation, the trapping of ionization-induced electrons and holes can result in the formation of charged defects, as in the case of yttria-stablized ZrO_2 [17]. These charged defects can affect defect structure, trapping rates of electrons and holes, defect formation and migration energies [18], and the kinetics of microstructure evolution, as in the faster grain growth observed in nanocrystalline cubic ZrO_2 under irradiation at cryogenic temperatures compared to higher temperatures [19]. In the case of ZnO, it has been shown that the diffusion mechanism for the oxygen interstitial is dependent on the defect charge state [20]. Other important properties, such as the luminescence quenching rates, may also be influenced by the charge state.

Recently, the charge states of defects in fluorite-structured oxides have gained increasing interest [21–24], but these studies are mainly focused on the formation of charged defects. At present, the stability and migration of charged oxygen interstitials in fluorite-structured oxides, including the migration energies and pathways, has not yet been well studied. In this study, we investigate charged oxygen interstitial formation and migration in ThO_2 and CeO_2. Here, CeO_2 is used as a surrogate material in place of PuO_2 for performance evaluation in harsh environments [25–27], because they share the same crystallographic structure, and several thermophysical and chemical properties. Our main aims are: (i) to gain fundamental insight into charged oxygen interstitial behavior in ThO_2 and CeO_2; (ii) to motivate experimental studies and provide support for their interpretation; and (iii) to provide physical parameters (e.g. defect configurations and energetics) for higher-scale modeling of fluorite-structured oxide fuels.

2. Computational details

All the calculations were carried out within the density functional theory (DFT) framework using the projector-augmented wave method, as implemented in the Vienna Ab Initio Simulation Package (VASP) [28]. The exchange-correlation effects were treated by the local-density approximation (LDA) in the Ceperley–Alder parametrization [29], with spin-polarized effects considered. Structural optimizations were carried out at variable cell shape and volume, as well as under the condition that the Hellmann–Feynman force on each atom should be smaller than 0.01 eV $Å^{-1}$. Computations were based on a $2 \times 2 \times 2$ supercell consisting of 96 atoms, with a $4 \times 4 \times 4$ k-point sampling in reciprocal space and a cutoff energy of 550 eV for the plane-wave basis set. Our previous work on oxygen vacancy formation and migration in CeO_2 has shown that the Hubbard U correction for the on-site Coulomb interaction must be taken into account due to the existence of Ce^{3+} [30], which is consistent with other theoretical studies on reduced ceria [31–33]. In this study, plus U correction calculations were also performed to investigate if the Hubbard U correction for Th 5*f* (or Ce 4*f*) electrons is necessary for ThO_2 (or CeO_2) containing an oxygen interstitial. In the LDA + U calculations, the method proposed by Dudarev et al. [34] was used, and effective U values of 4 eV for Th and 6 eV for Ce, obtained from our previous study [30], were employed. The minimum-energy pathways for oxygen interstitial migration in ThO_2 and CeO_2 were investigated by the climbing-image nudged elastic band method [35].

3. Background

For systems with charged defects, spurious electrostatic and elastic interactions between the defect and its periodic images can be large, and the system converges very slowly with respect to the supercell size. To correct the electrostatic effects on the formation energies of charged defects, several approaches have been proposed [36–40]. Here, the screened Madelung-like lattice energy of point defects proposed by Leslie and Gillan [39] is employed, i.e. $E_{corr} = \frac{q^2 \alpha_M}{2\varepsilon L}$, where $L = \Omega^{-1/3}$ is the linear supercell dimension, Ω is the volume of the supercell, ε is the dielectric constant and α_M is the appropriate Madelung constant for the respective supercell geometry. Another correction for the charged system is the potential alignment, ΔV, which is used to estimate the valence band maximum (VBM) of the defective supercell. The ΔV value is obtained by the averaged difference between the local potentials far from the defect in the defective supercell and the corresponding ones in the perfect supercell [41].The defect formation energies are defined by $E_f = E_{def} - E_{perfect} + n_i \mu_O + q\left(E_F + E_{VBM}^{bulk} + \Delta V\right) + E_{corr}$ [42], where E_{def} and $E_{perfect}$ are the total energies of the supercell with and without a defect, respectively, n_i is the number of atoms being removed or added, and μ_O is the chemical potential of O. The value of μ_O is obtained under O-rich or M-rich (M = Th or Ce) conditions. Under O-rich conditions, the experimental chemical potential of O (μ_O) from the free O_2 molecule is employed [43]; under M-rich conditions, μ_O is obtained by $\mu_{MO_2} = \mu_M + 2\mu_O$, where μ_{MO_2} and μ_M are calculated from the total energies of bulk MO_2 and M, respectively. The term in parentheses is the electronic chemical potential that represents the change in energy associated with charged defects. In this term, E_F is the Fermi level in the bulk with reference to the VBM, which can vary within the band gap, E_{VBM}^{bulk} is the energy of the bulk VBM, and ΔV is the electrostatic potential alignment of the defective structure. The additional charge q is set by adding or removing electrons from the supercell. The last term, E_{corr}, is the first-order Madelung correction energy.

4. Results and discussion

4.1. Comparison of LDA + U and LDA calculations for oxygen interstitial formation

Pulsed neutron diffraction experiments performed on a nanoscale powder of CeO_2 by Mamontov and Egami

revealed oxygen interstitial occupation at the octahedral site of the CeO_2 [3]. Our *ab initio* molecular dynamics simulation of low-energy recoil events in ThO_2, CeO_2 and ZrO_2 also showed that, under low-energy irradiation, oxygen interstitials generally occupy the octahedral site ($O_{octa.}$) [2]. In addition to this octahedral site, the O–O⟨1 1 0⟩ split interstitial (O_{split}) is also a possible stable defect. These two oxygen interstitial (O_{int}) configurations, as shown in Fig. 1, are considered as the initial configurations in the present work to study charged oxygen interstitial incorporation and migration in ThO_2 and CeO_2.

To investigate the effect of considering the Hubbard U correction for Th 5*f* and Ce 4*f* electrons on the stability of oxygen interstitial in ThO_2 and CeO_2, the formation energies for the octahedral oxygen interstitials with charge states varying from 2− to 2+ are first calculated by both LDA and LDA + U methods. The variation of the formation energies as a function of the Fermi level with different charge states in both oxides is shown in Fig. 2. Under both O-rich and Th-rich conditions, it appears that consideration of the Hubbard U correction for the Th 5*f* electrons has only a small influence on the interstitial formation energies in ThO_2. In CeO_2, the plus U correction for the Ce 4*f* electrons affects the $O^{2-}_{octa.}$ interstitial formation under both O- and Ce-rich conditions, which narrows the stability region by ~0.25 eV. The density of state distribution for the $O^{2-}_{octa.}$ interstitial in CeO_2 and ThO_2 obtained by both methods is further analyzed, as illustrated in Fig. 3. It turns out that the standard LDA describes well the splitting of Ce 4*f* and Th 5*f* electrons into occupied and unoccupied states, and LDA + U only pushes the unoccupied states to a higher energy level. This is different from the case of oxygen vacancy formation in CeO_2 [30], for which the Hubbard U correction for the on-site Coulomb interaction is necessary to localize the Ce 4*f* electrons. For the split oxygen interstitials in both oxides, the LDA + U and LDA methods also predict very similar formation energies, and the *f* electrons are well localized under standard LDA. Consequently, the plus U correction is not considered in the subsequent calculations, and all the following results and discussions are based on standard LDA calculations.

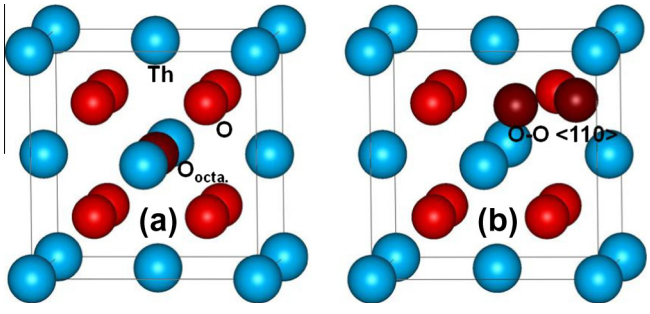


Fig. 1. Schematic view of (a) octahedral ($O_{octa.}$) and (b) split (O_{split}) oxygen interstitial in ThO_2.

4.2. Stability of oxygen interstitials

The defect formation energies for octahedral and split oxygen interstitials in ThO_2 and CeO_2 are presented in Fig. 4a and b, respectively. Because oxygen interstitials are more favorable under O-rich conditions than Th-rich conditions, only the O-rich condition is considered here. In ThO_2, it is found that the O^{+}_{split}, O^{0}_{split}, $O^{-}_{octa.}$ and $O^{2-}_{octa.}$ are the lowest-energy configurations in the Fermi energy ranges 0–0.32, 0.32–1.35, 1.35–2.73 and 2.73–4.04 eV, respectively. The geometrical configurations for the O^{+}_{split}, O^{0}_{split} and $O^{2-}_{octa.}$ interstitials in ThO_2 are shown in Fig. 5. Our calculations indicate that the stability of oxygen interstitial depends on the charge state, Fermi level and oxygen interstitial configuration. The stability of oxygen interstitial in ThO_2 is found to be different from that in UO_2 [21,22,44], for which the O^{2-} interstitial is more stable over most of the Fermi energy range. Such a discrepancy may be caused by the different interactions between the oxygen interstitial and its neighbors.

In CeO_2, the O^{+}_{split}, O^{0}_{split} and $O^{2-}_{octa.}$ interstitials are the lowest-energy configurations in the Fermi energy ranges 0–0.39, 0.39–1.76 and 1.76–2.17 eV, respectively, as shown in Fig. 4b. Similar to the case of ThO_2, the O^{+}_{split} interstitial is preferred under *p*-type conditions, where the Fermi level is very close to the valence band maximum, and $O^{2-}_{octa.}$ is favored under *n*-type conditions, i.e. as the Fermi level approaches the conduction band minimum. These results suggest that, in both ThO_2 and CeO_2, the oxygen interstitial is likely to lose electrons under *p*-type conditions and gain electrons under *n*-type conditions, behaving as a donor and acceptor defect, respectively. Erhart et al. [45] also suggested the donor- and acceptor-like characters of the oxygen interstitial atoms in ZnO. It should be pointed out that the singly positive split interstitials are only stable when the Fermi level is very close to the valence band maximum. Bader charge analysis shows that these singly positive split interstitials donate electrons to all of their first-neighboring cations, resulting in a positive charge state of the interstitials. Another finding is that the energy difference between the octahedral and split interstitials with a specific charge state is much smaller in ThO_2, and that $O^{-}_{octa.}$ is one of the lowest-energy configurations within the Fermi energy range in ThO_2 but not in CeO_2. This suggests despite the fact that ThO_2 and CeO_2 have the same fluorite-type structure, the size and electronic configuration of the cations may significantly affect the oxygen interstitial stability.

To study the effect of charge states on the interatomic structures, the structural properties for oxygen interstitials with different charge states in both oxides have been further analyzed, as summarized in Table 1. For the octahedral interstitials, the distance between the oxygen interstitial and its neighboring lattice oxygens increases as *q* increases from 2− to 2+. This is because in MO_2 the lattice O ions are negatively charged, and the introduction of negatively or positively charged O_{int} will push the nearest lattice oxygens away due to the repulsive interaction, or pull the

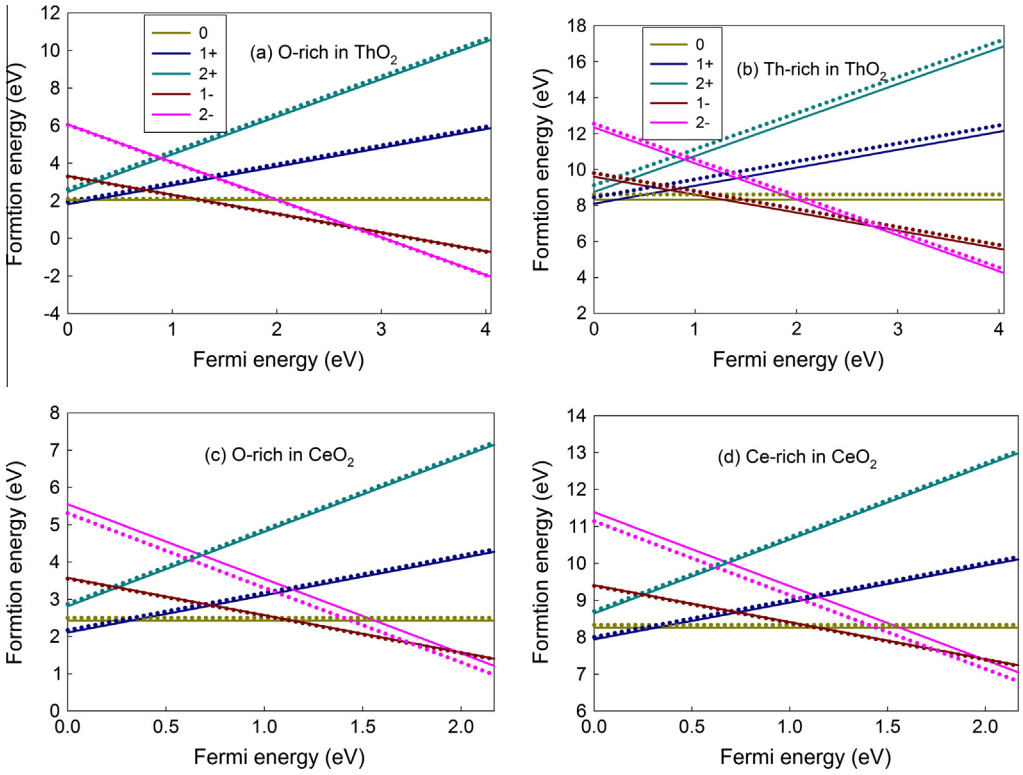


Fig. 2. Variation of oxygen interstitial formation energies with Fermi level in (a and b) ThO_2 and (c and d) CeO_2 under O- and M-rich (M = Th or Ce) conditions. Solid and dotted lines correspond to the results obtained from LDA + U and LDA calculations, respectively.

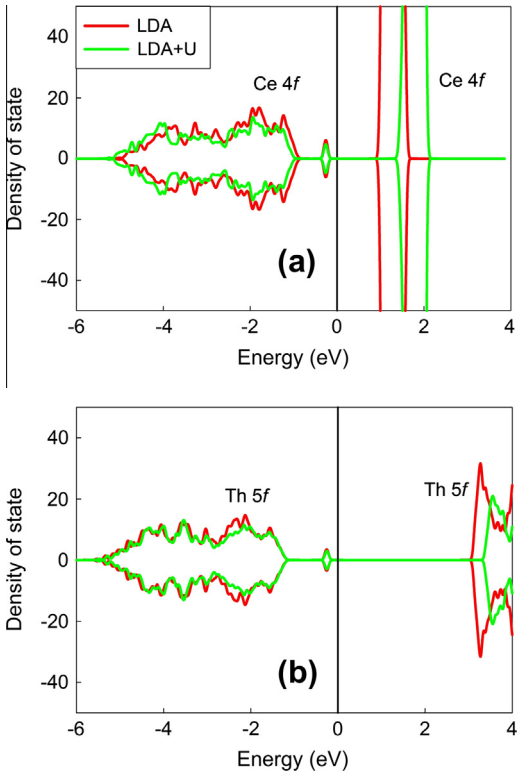


Fig. 3. Density of state for $O^{2-}_{octa.}$ interstitial in (a) CeO_2 and (b) ThO_2 obtained by LDA and LDA + U calculations. The Fermi level is set at the zero of the energy.

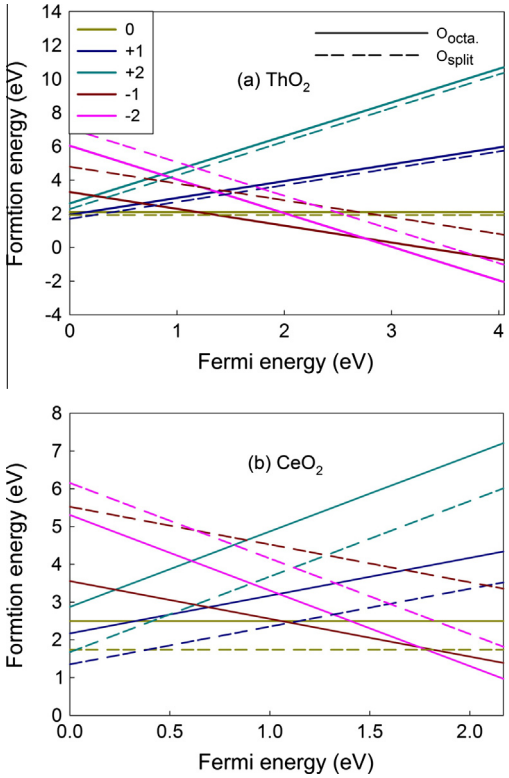


Fig. 4. Oxygen interstitial formation energies as a function of the Fermi energy in (a) ThO_2 and (b) CeO_2. Solid and dashed lines represent octahedral and split interstitials, respectively.

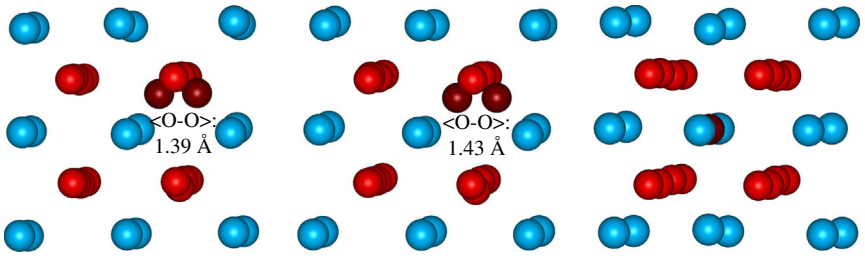


Fig. 5. Schematic view of geometrical configuration for (a) O^{+}_{split}, (b) O^{0}_{split} and (c) $O^{2-}_{octa.}$ in ThO_2.

Table 1
Geometrical properties for oxygen interstitial in ThO_2 and CeO_2: $d\langle O_{int}–O_{int}\rangle$ is the $\langle O–O\rangle$ bond length of the split interstitial; $d\langle O_{int}–O_{lat}\rangle$ is the distance between the interstitial and its neighboring lattice oxygens; and ΔV is the defect formation volume relative to the perfect volume (V_0).

	ThO_2			CeO_2		
	$d\langle O_{int}–O_{int}\rangle$ (Å)	$d\langle O_{int}–O_{lat}\rangle$ (Å)	$\Delta V/V_0$ (%)	$d\langle O_{int}–O_{int}\rangle$ (Å)	$d\langle O_{int}–O_{lat}\rangle$ (Å)	$\Delta V/V_0$ (%)
O^{2+}_{split}	1.36	2.38	−0.28	1.31	2.28	−0.41
O^{+}_{split}	1.38	2.41	0.22	1.35	2.35	0.20
O^{0}_{split}	1.43	2.47	0.74	1.39	2.41	0.93
O^{-}_{split}	1.99	2.43	1.24	1.40	2.42	1.78
O^{2-}_{split}	2.43	2.42	1.76	2.35	2.35	2.23
$O^{2+}_{octa.}$	–	2.44	−0.90	–	2.36	−1.12
$O^{+}_{octa.}$	–	2.45	−0.39	–	2.37	−0.46
$O^{0}_{octa.}$	–	2.47	0.14	–	2.40	0.23
$O^{-}_{octa.}$	–	2.51	0.69	–	2.43	0.98
$O^{2-}_{octa.}$	–	2.54	1.26	–	2.47	1.87

nearest lattice oxygens closer due to the attractive interaction, respectively. Consequently, positive oxygen interstitials cause volume compression, and neutral or negative oxygen interstitials cause volume expansion. In the case of split interstitials, the $\langle O_{int}–O_{int}\rangle$ distance increases as q varies from 2+ to 2−; however, the distances between either O_{int} of the split interstitials and their neighboring lattice oxygens do not exhibit a singular change. Furthermore, the introduction of the O^{2+}_{split} into the system causes volume compression, and the split interstitials with other charge states cause volume swelling. Obviously, the charge states have considerable effects on the geometrical structures for both split and octahedral interstitials.

For some split interstitials in ThO_2 and CeO_2, it is noted that the $\langle O_{int}–O_{int}\rangle$ bond distance is close to that of a free O_2 molecule, i.e. 1.21 Å. However, the O–O split interstitial interacts with its neighboring atoms, and does not form an O_2 molecule. Fig. 6 shows the charge density distribution for the O^{+}_{split} interstitial in ThO_2. The $\langle Th–O_1\rangle$ and $\langle Th–O_2\rangle$ bond lengths are each 2.46 Å. Each O_{int} in the split interstitial interacts with two lattice oxygens at bond distances of 2.41 and 3.38 Å, respectively. The average Bader charge of the two oxygens in the split interstitial is +0.58 |e|, as compared with the charge of ±0.16 |e| in a free O_2, suggesting that the two oxygens in the split interstitial do not form a free molecule. Our theoretical prediction of the structural and electronic properties of the O^{+}_{split} provides fundamental information for the design and interpretation of planned low-temperature irradiation experiments in these fluorite-structured oxides.

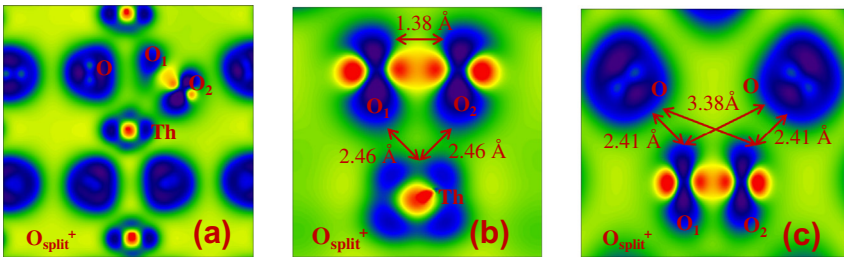


Fig. 6. Charge density distribution for O^{+}_{split} in ThO_2 projected on planes (a) containing interstitials and their first neighboring cations and anions, (b) containing interstitials and their first neighboring cations and (c) containing interstitials and their first neighboring anions.

Table 2
Energy barrier for oxygen interstitial migration in ThO_2 and CeO_2.

	ThO_2	CeO_2
O^{+}_{split}	0.13	0.56
O^{0}_{split}	0.32	0.76
$O^{-}_{octa.}$	0.98	–
$O^{2-}_{octa.}$	1.04	0.80

4.3. Oxygen interstitial migration

The calculated energy barriers for oxygen interstitial migration in ThO_2 and CeO_2 are given in Table 2. In both oxides, the O^{+}_{split} is the most mobile, with migration energies of 0.13 and 0.56 eV in ThO_2 and CeO_2, respectively. The O^{0}_{split} interstitials, which also have a high degree of stability, have only slightly higher migration energies. These results indicate that the oxygen split interstitial is more mobile in ThO_2 than in CeO_2. The activation energy for oxygen self-diffusion in ThO_2, which occurs via oxygen vacancies, has been determined experimentally [1,14]; however, direct experimental measurement of oxygen vacancy and interstitial migration energies have not been reported.

As shown in Fig. 7, the saddle-point for O^{+}_{split} migration corresponds to the oxygen interstitial occupying an octahedral site $\left(O^{+}_{octa.}\right)$, which was shown to be a metastable state in Fig. 4. The distance between the O_{int} and its neighboring Th atoms is 2.79 Å, which is 0.33 Å larger than that in the starting configuration, indicating that the covalency for the ⟨Th–O⟩ bond at the saddle-point is weakened. A number of theoretical studies have been performed on neutral oxygen interstitial diffusion in UO_2 [11,12,46,47]. In these calculations for UO_2, the direct interstitial mechanism, in which the oxygen interstitial diffuses directly from an octahedral site to another one along the ⟨1 1 0⟩ direction, passing through the bridge center between two neighboring lattice O atoms, and the interstitialcy mechanism, in which the migrating defect replaces one lattice atom, and the lattice atom then becomes the migrating species, have been considered [46,48]. Although different migration barrier values were obtained, these calculations suggested that the interstitialcy mechanism in UO_2 is more energetically preferable than the direct interstitial mechanism. For ThO_2 and CeO_2, the interstitialcy mechanism is also found to be more favorable than the direct interstitial mechanism. The case of $O^{2-}_{octa.}$ interstitial migration in ThO_2 via the interstitialcy mechanism is illustrated in Fig. 7, where the saddle-point corresponds to the O^{2-}_{split} configuration. However, the energy barrier is significantly higher than that for O^{+}_{split} interstitial migration.

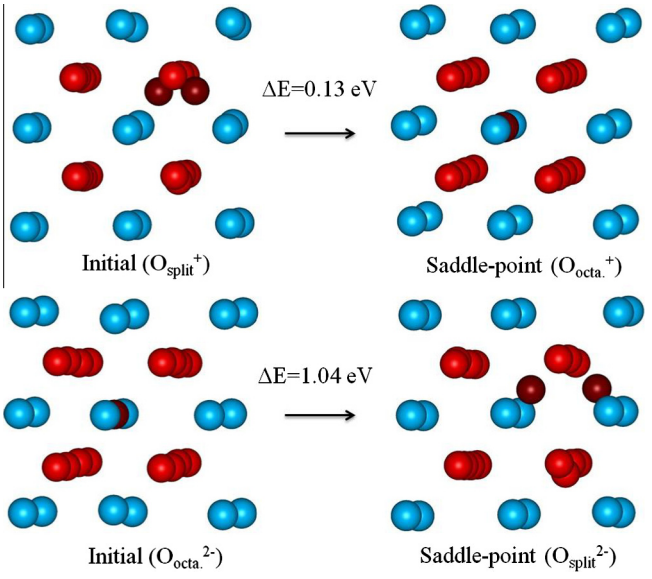


Fig. 7. Migration pathway for O^{0}_{split} and $O^{2-}_{octa.}$ in ThO_2.

Our calculations suggest that oxygen interstitial migration in ThO_2 and CeO_2 is influenced by the charge state, configuration of the oxygen interstitial, and the composition of host material. In ZnO, Erhart and Albe [20] also found the O^{2-} interstitial migrates via a different mechanism than the O^{0} and O^{2+} interstitials. The fact that the migration barrier and pathway are strongly dependent on the charge state in some oxides suggests the possibility of using an applied electric field to affect the diffusion and to control defect concentrations [48]. The migration behavior of the oxygen interstitial in ThO_2 and CeO_2 generally exhibit similar characteristics, whereas the size and electronic configuration of the cations still influence the migration barriers, resulting in lower oxygen interstitial mobility in CeO_2. It is expected that these results may provide some new insights for future theoretical and experimental studies of point defects behavior in fluorite-structured oxides.

5. Conclusions

In summary, an *ab initio* method based on DFT has been employed to study the stability and migration of charged oxygen interstitial in ThO_2 and CeO_2. It is shown that the Fermi level has a significant influence on the stability of oxygen interstitial. In ThO_2, the O^{+}_{split}, O^{0}_{split}, $O^{-}_{octa.}$ and $O^{2-}_{octa.}$ are the lowest-energy configurations in the Fermi energy ranges 0–0.32, 0.32–1.35, 1.35–2.73 and 2.73–4.04 eV, respectively. In the case of CeO_2, the O^{+}_{split}, O^{0}_{split} and $O^{2-}_{octa.}$ interstitials are the lowest-energy configurations in the Fermi energy ranges 0–0.39, 0.39–1.76 and 1.76–2.17 eV, respectively. These results suggest that oxygen interstitials behave as donor- and acceptor-like defects under *p*-type and *n*-type conditions, respectively. Split and octahedral interstitials with different charge states can cause local volume compression or swelling. Although the ⟨O–O⟩ bond length of split interstitial is close to that of a free O_2, the two oxygens interact with their neighbors rather than forming a free O_2 molecule.

In both oxides, the O^{+}_{split} migration has the lowest migration barrier, for which the saddle-point corresponds to the oxygen interstitial occupying an octahedral site. The lowest migration energies are determined to be 0.13 eV in ThO_2 and 0.56 eV in CeO_2, indicative of a higher mobility for the oxygen interstitial in ThO_2. Our calculations suggest

that the oxygen interstitial behavior in ThO_2 and CeO_2 is influenced by the charge state, interstitial configuration and composition of the host material. The stability and migration of the oxygen interstitial in both oxides generally exhibit similar characteristics; however, the migration barriers are significantly affected by cation size and electronic configuration.

Acknowledgements

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